

Animal dataset classification tasks*

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Abstract—Image classification is a popular topic in computer vision and is widely used in various fields such as medical diagnosis, wildlife monitoring, e-commerce and autonomous driving. For this project, I chose the dataset provided by Hugging Face: mertcobanov/animals dataset, a multi-class dataset containing labeled animal images. The image data is first collected and data analysis is performed on the dataset. Then some computer vision data augmentation methods are used. In addition, this project attempts to use a variety of classification methods, including methods such as Support Vector Machines (SVMs) and Convolutional Neural Networks (CNNs), and tries to use the model provided by Hugging Face.

Index Terms—Image Classification, Computer Vision, Animal Dataset, Data Analysis

I. INTRODUCTION

A. Background and motivation

Image classification, a fundamental task in computer vision, plays an important role in many domains. Its wide range of applications has stimulated interest in developing efficient and accurate classification models. In this project, my primary motivation is to deepen my understanding of image classification techniques, from data preprocessing and augmentation to model training and evaluation. To achieve this, I have selected the mertcobanov/animals dataset, a multi-class dataset available on Hugging Face. The dataset contains labeled animal images, providing an opportunity to explore the challenges and opportunities associated with the classification task.

B. Project goal

The goal of this project is to use Python to perform data analysis on the mertcobanov/animals dataset and to classify animal images in the dataset, with a focus on evaluating and comparing the performance of various machine learning models. This includes applying data augmentation techniques (e.g., Rotate, Flip, Cutmix, Mixup, Autoaugment) to increase the diversity of the dataset and exploring traditional machine learning algorithms such as Support Vector Machines (SVMs) and deep learning methods such as Convolutional Neural Networks (CNNs).

C. Review of existing literature

SVM have been widely used for image classification tasks. Studies have shown that combining SVM with feature extraction methods, such as the use of pre-trained CNNs such as ResNet, can achieve competitive performance. (Niu et al.,

2012) proposed a hybrid model that integrates the synergy of two advanced classifiers: CNN and SVM. In this model, CNN acts as a trainable feature extractor and SVM acts as a recognizer. This hybrid model automatically extracts features from the original image and generates predictions. Comparison with other studies shows that this fusion achieves better results. (Ozaltin et al., 2022) proposes a novel CNN to automatically extract features from images and combines it with SVM, the classification accuracy is greatly improved. The framework can help clinicians to perform ECG type recognition.

Convolutional Neural Networks (CNN) revolutionized image classification with architectures like LeNet, AlexNet, and ResNet. The introduction of ResNet (He et al., 2016) solved the problem of gradient vanishing in deep networks by introducing residual connectivity.

Data augmentation techniques such as Mixup (Zhang et al., 2018), which generates new training samples by mixing two images and their corresponding labels in a certain ratio (pixel-level mixing). CutMix (Yun et al., 2019), which generates new training samples by cutting and pasting a portion of one image onto another (block-level mixing) while mixing labels. AutoAugment (Cubuk et al., 2019) automatically learns an optimal set of data augmentation strategies. It aims to augment the diversity of the data by searching for combinations of augmentation strategies to maximize the performance of the dataset.

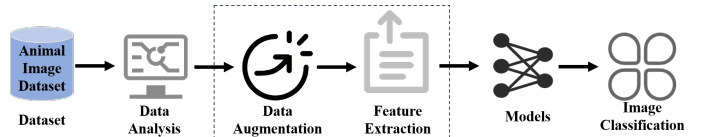


Fig. 1. Flowchart of the project.

II. METHOD

A. Problem formulation

The dataset contains 5,400 labeled images, evenly distributed across all 90 classes, with no class imbalance issues. The goal of this project is to classify images of animals from the mertcobanov/animals dataset into one of 90 categories. The input is an RGB image with varying dimensions and images are resized to a consistent resolution of 224×224 pixels. The output is a single predicted class label corresponding to the animal depicted in the image. The project uses a combination of CNN and SVM. Feature extraction using a pre-trained ResNet-18 model allows the use of Support Vector

Machines (SVM) for efficient classification. For more complex approaches, Classical CNN and Hugging Face’s pre-trained models, such as ResNet-50. The flow of this project is shown in Figure 1.

B. Methodologies used to solve the problem

The first step is dataset preprocessing, where images are resized to 224×224 pixels and normalized for consistency across the dataset. To enhance model generalization, data augmentation techniques are used. The dataset is then classified using the SVM algorithm. Besides, feature extraction plays a critical role in addressing computational constraints. A pre-trained ResNet-18 model is used to extract meaningful features, which are then input into an SVM classifier. For deep learning models, CNNs are trained directly on the dataset, but due to local CPU limitations, training epochs are limited. Pre-trained models from Hugging Face (e.g., ResNet-50) were also used for this multi-classification task.

III. RESULTS

A. Data pipeline or model set up

The data pipeline begins with loading the mertcobanov/animals dataset. Preprocessing ensures that the data is consistent and ready for model training. Each image is resized to a fixed resolution of 224×224 pixels to standardize inputs. Data augmentation is applied to increase variability and improve model generalization. Methods include basic transformations such as rotate, flip and crop, and advanced techniques such as Cutmix, Mixup and AutoAugment. The dataset is then split into training, testing sets (e.g., 80:20 ratio) to evaluate model performance. For traditional machine learning approaches, features are extracted using a pre-trained ResNet-18 model to reduce input dimensionality while preserving key characteristics.

B. Figures that present numerical simulation results

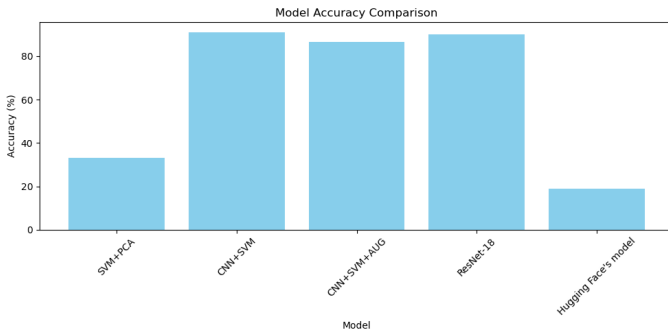


Fig. 2. Model Accuracy Comparison.

C. Interpretation of the results

Comparison of model accuracy and execution time is shown in Table 1. PCA reduces the dimensionality of the dataset and thus increases the computational efficiency. So the running

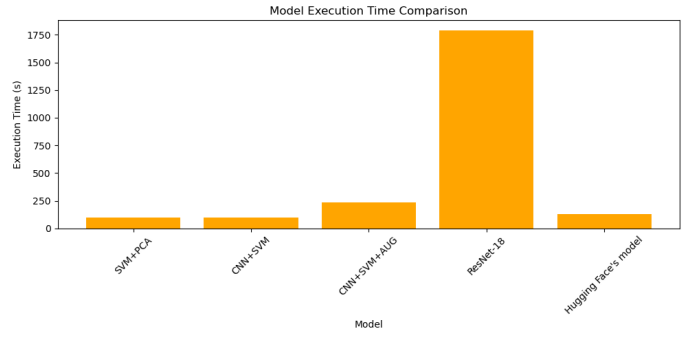


Fig. 3. Model Execution Time Comparison.

TABLE I
MODEL ACCURACY AND EXECUTION TIME COMPARISON

Model	Accuracy (%)	Execution Time (s)
SVM+PCA	33%	99.83
CNN+SVM	91.11%	99.94
CNN+SVM+AUG	86.67%	237.16
ResNet-18	90.09%	1791.82
Hugging Face's model	19%	131.96

time of the model is shorter. However, PCA may omit important information that is critical for accurate classification, resulting in limited accuracy. The synergy between the feature extraction of CNNs and the classification capabilities of SVMs results in excellent performance, highlighting the advantages of this combined approach. Also the execution time of the model is shorter. However the fact that the data augmentation did not improve the accuracy may be due to the fact that the quality of the dataset is already high, and the data augmentation may even introduce additional noise, leading to a decrease in the accuracy. The best augmentation method is the basic transformation, which is listed in Table 1. ResNet-18 is relatively accurate, but very computationally expensive compared to other models, 5 epochs took 1791.82 seconds. The pre-trained ResNet-50 performed poorly with only 19% accuracy. The ResNet-50 pre-trained model benefits from features learned on large datasets. However, these features may be less applicable to the animal dataset in this project, resulting in a lower accuracy.

IV. CONCLUSION

CNN + SVM showed the best performance by effectively combining feature extraction and classification. Data augmentation is promising but less accurate, suggesting the need for careful design and validation based on the dataset. ResNet-18 is also highly accurate, but computationally expensive, and the use of GPUs can greatly reduce execution time. Hugging Face’s model ResNet-50 performed poorly, suggesting that the training data has a large impact on the model’s performance, and that specialization is required for specific tasks. For specific tasks that require specialized training models, data is critical.